

How Important is Immersion for Learning in Computer Science Replugged Games?

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Figure 1: Navigating through the immersive ‘Pengu’s Treasure Hunt’ activity in Virtual Reality, 2019.

ABSTRACT

By following the idea of not using computers at all, Computer Science Unplugged has set the course for many interactive, social, and hands-on activities dealing with concepts and problems of Computer Science Education. Through recent developments in immersive media, technologies like Virtual and Augmented Reality could enhance such activities or even enable new ones. When providing immersive educational media that induce a sense of presence in the virtual environment, the illusion of unmediated learning experience can be delivered. Hence, the concept of “Computer Science Replugged” can benefit from the affordances that Computer Science Unplugged thrives on while facilitating or enabling activities that might be impossible, dangerous, or expensive to carry out in reality. This paper presents three concepts from Computer Science Education that have been modeled as 3-D immersive educational virtual environments: components of a computer, asymmetric encryption/decryption, and finite state machines. To get a first impression of the effectiveness of these approaches and in order to

determine the importance of the level of immersion for the learning process, a study with 78 participants was conducted in which the software was tested on different devices. All activities were found to be significantly effective with regards to the pre- and post-tests. When analyzing these results on the basis of comparing the least immersive setting (laptop) with the most immersive setting (head-mounted-display), the findings indicate different effects with effect sizes between $\beta = -.17$ and $\beta = .41$. This raises two questions: “Which topics from Computer Science Education can benefit from immersive technology?” and “What are the opportunities and challenges of the didactical design of Computer Science Replugged activities?” that have to be addressed in further research.

CCS CONCEPTS

• **Social and professional topics** → **K-12 education**; • **Human-centered computing** → *Virtual reality*; HCI theory, concepts and models.

KEYWORDS

Computer Science Unplugged, Immersive Learning, Virtual Reality, Technology-Enhanced Learning

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1 INTRODUCTION

Computer Science Unplugged activities try to make various contents from Computer Science (CS) easy and fun to learn by using game-like settings, metaphors, and hands-on interaction. Computer Science Unplugged “takes the unusual approach of exposing children to the great ideas of Computer Science without using computers” [2]. While a key principle of Computer Science Unplugged is the development of computer-independent teaching methods for CS, other principles for designing activities for teaching and learning CS are: focus on CS concepts rather than programming, making the activities kinaesthetic through involving teamwork, making the activities fun and engaging, low cost, release under a creative common license, making the activities gender neutral and rather focus on cooperation than on individualistic approaches, and adding a sense of story to the activities [2]. In order to modify, re-create, or enhance several CS Unplugged activities, virtual games or simulations have been developed in the recent years in order to teach about these concepts. Educational Virtual Environments (EVEs) are safe, less prone to failure than activities carried out in real environments, and easy to use and re-use once the required hardware is procured. By following the *Computer Science Replugged* approach, which integrates technology into CS Education to provide hands-on virtual activities in immersive environments, this paper explores the question whether different levels of immersion in EVEs for CS Education influence the level of experienced presence and the learning outcomes from using the environments.

2 BEING THERE: VIRTUALIZING COMPUTER SCIENCE UNPLUGGED

The key characteristics of Virtual Realities (VRs), environments solely consisting of virtual components [13], are immersion and presence. Immersion can be seen as a quantifiable description of the used technology [15], while a popular understanding of presence is the perception of non-mediation [10], as the feeling of ‘being there’ [7]. Presence is thought of to be a unique characteristic of Educational Virtual Environments [12]. Studies show that the level of immersion has a strong impact on presence [9, 11, 14].

It has to be noted that the discussion of immersion and presence is way more complex than presented here. We decided in favor for this distinction as it allows to separate the supply-side of the educational technology from the subjective use side (the perception and learning activities) of the learner [5]. Doing so, immersion becomes part of the methodological decisions of the teacher who decides which software is provided with which hardware at which point of the learning process; presence mediates the student’s learning activities in terms of perception and interpretation. Of course, the provided level of immersion can predict the sense of presence to a certain level, though Dengel and Mägdefrau note that person-specific factors like motivation, cognition, and emotion influence the learner’s sense of presence as well [6].

The idea of *Computer Science Replugged* [blinded for review] makes use of these factors influencing learning in immersive educational virtual environments: it provides hands-on activities combined with immersive technology that simulate a perception of non-mediation. Doing so, the idea of being independent from using computers as an educational medium can be maintained, following

the Computer Science Unplugged approach: The device is only used to simulate the experience (while the technology itself disappears in the experience) rather than being an instrument for teaching and learning (as it would be the case for coding tasks on a desktop PC).

In section 3, we will present three approaches that put the Computer Science Replugged approach to practice. In order to evaluate the underlying ideas of combining metaphorical and schematizing visualizations with hands-on interaction tasks, the first research question of this paper asks if the presented EVEs following the concept of Computer Science Replugged are able to induce learning processes in terms of measurable learning outcomes.

While systems including head- and body-tracking are considered to be greatly immersive, they are often way more expensive than low-immersive systems (e.g. mobile VR devices) and non-immersive systems (e.g. laptops, desktop PCs). Therefore, it is of great interest how learning processes and outcomes benefit of the provided level of immersion when reasoning about purchasing immersive hardware for educational purposes. Hence, the second research question of this paper is concerned with comparing non-immersive and immersive technologies regarding learning outcomes. As it can be assumed that different levels of immersion contribute to learning differently, the third research question asks if there is a positive relation between the level of immersion and the learning outcomes.

3 COMPUTER SCIENCE REPLUGGED ACTIVITIES

3.1 Components of a Computer

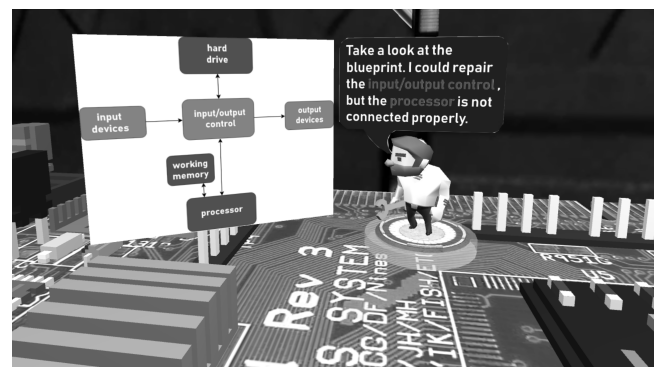


Figure 2: Bill’s Computer Workshop

The learning contents for the first EVE derive from the bavarian curriculum for the twelfth grade. The learning objectives were: 1) The students can name the basic components of a computer (mainboard/motherboard, primary memory/RAM, secondary memory/hard drive, processor/CPU, input devices, output devices) and their functions; 2) The students can allocate these components in a relational scheme. In our immersive EVE “Bill’s Computer Workshop”, the students start their journey in a small home office with a computer. When trying to start the computer, the monitor displays an error. In order to fix it, the player has to enter the computer. Inside the computer case, the student encounters Bill the Mechanic, who tries to repair the computer’s components but he needs the

player's help. After explaining the function of the motherboard, Bill shows the player a schematic plan of the components and their relations, adapted from a German schoolbook [8]). Then, the student has to find, pick up, and insert the processor in the correct slot. Afterwards, Bill explains what the processor is used for and instructs the user to get the RAM bars. After finding and bringing them back to their slots, Bill explains further how the processor and the primary memory work together. As a last task, Bill assigns the player to reconnect the secondary memory in order to fix the computer's long-term storage. In the end, Bill sums up the components of the computer again and then runs a few tests, showing how the components work together to fulfill tasks like processing keyboard inputs or playing and saving a video file. When getting back into the home office, a displayed smile on the monitor indicates that their computer is working again.

All controls for the laptop, the mobile VR, and for the HTC Vive were gaze-based. Locomotion was realized via teleportation between fixed teleporter platforms and the interaction with the environment (picking up components, pressing buttons, dialogues, etc.) was realized via point-and-click (also gaze-based).

3.2 Asymmetric Encryption

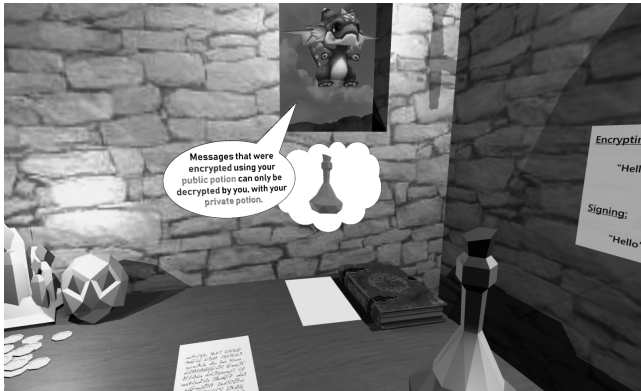


Figure 3: Fluxi's Cryptic Potions

The second EVE, *Fluxi's Cryptic Potions*, tried to combine the original Computer Science Unplugged activity [16], where the students communicate in an unknown network and send each other messages (or chocolate bars) with the mixing colours idea [1] where students mix colours in order to get a secret colour (representing a secret key for encrypting and decrypting messages). Our medieval setting takes the player into a castle chamber where he/she encounters Fluxi, the player's carrier dragon, who delivers messages to and from the student. First, the player receives a letter from his/her friend Nikolay who asks the player if he/she will be at Sir Dance-A-Lot's (the metaphorical man-in-the-middle) party this evening. After reading the letter, Fluxi asks the player to reply to Nikolay by telling him that he/she wants to attend but has not received an invitation yet. After Fluxi brings the letter to the post office (simulating the network structure), he returns with an encrypted answer from Nikolay. Fluxi explains that the post office provides each participant of the network with two cryptic potions: a private and a public potion. While all potions cipher messages, the

encryption can only be reversed by using the corresponding other potion. The public potion of each participant is stored publicly in the post office and everyone can get a copy of it. In contrast, the recipe of the private potion is secret and only known by the user. After explaining the benefits of this asymmetric encryption process, he instructs the user to decrypt Nikolay's message by using his/her own private potion. Nikolay informs the player that Princess Isolde, Prince Charming, and Fluxi's aunt Gertrude will be at the party as well. He notes that the player always wanted to dance with one of them and that he/she should write a letter to the person of interest. But Nikolay also warns the player that Sir Dance-A-Lot wants to dance with all of them which is why the communication should be kept a secret and, therefore, all messages should be encrypted. The player writes a new, encrypted letter to Nikolay (with Nikolay's public potion) and agrees that it would be a good idea to encrypt the messages. After delivering this letter to the post office, Fluxi returns with the invitation of Sir Dance-A-Lot. The invitation seems to be ciphered as well, and Fluxi explains that Sir Dance-A-Lot signed the invitation so that everyone knows that the message must be from him. After reasoning why this process is secure (in terms of authentication), the player adds Sir Dance-A-Lot's public potion to the letter in order to decipher it. The player writes a message saying that he/she will attend the party and signs the letter with his own private potion. After delivering the letter, Fluxi asks the player who it is he/she wants to dance with. As the dialogues and letters are quite similar and the tasks stay the same, we will present the scenario for a player who chose Prince Charming. The player writes a message to Prince Charming asking him for a dance. Fluxi gets the prince's public potion from the post office and explains that this potion can encrypt letters for Prince Charming and decipher signed letters from him as well. The player encrypts the message with Prince Charming's public potion (to avoid Sir Dance-A-Lot reading it) and gives the letter to Fluxi. After returning from the post office, Fluxi gives the player a signed response from Prince Charming: He does not believe the player's authenticity as the player encrypted the message for Prince Charming, but did not sign it. Hence, the player has to rewrite his/her letter, encrypt it with Prince Charming's public potion and sign it with his/her own private potion. After resending the letter, Fluxi returns with a signed and encrypted reply from Prince Charming, telling the player that the prince waited an eternity for this question and would be glad to dance with him/her.

Similar to the interaction in *Bill's Computer Workshop*, the controls in *Fluxi's Cryptic Potions* were gaze-based via point-and-click. The player could pick up potions and letters, write new messages, and talk to the dragon. In contrast to the other EVEs, the player had no possibility to move or teleport, resulting in him/her staying in the same room all the time. In all technological settings, the player sat on a stool, simulating the same position as in the EVE.

3.3 Finite State Machines

The third EVE, *Pengu's Treasure Hunt*, focuses the topic of finite state machines. It is orientated on the original Computer Science Unplugged activity [3]: Islands represent states, the treasure island(s) represents the final state(s), boats represent the inputs, and the boat routes represent the transition functions. The island worlds were

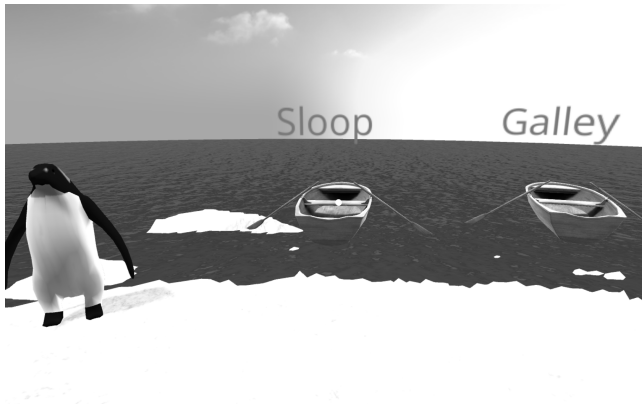


Figure 4: Pengu's Treasure Hunt

created using the finite state machine Virtual Reality converter presented in [4]. When starting the game, the learner lands on the main island and encounters Pengu, the fearsome pirate, who explains that the player can find his treasures by completing several island worlds. In order to open the treasures on the respective treasure island, the player has to find key parts on the different routes. The treasure chest cannot be opened unless all key parts (and therefore all routes between the islands, resulting in a complete finite state machine) are found. A map shows the player which islands and routes were found, as well as the remaining key parts. In the first world, consisting of three islands and two different boats, only one boat leads to the next island, the other one brings the user back to the same island. After finding all six routes, the player can open the treasure chest and returns to the main island. Pengu congratulates the player and introduces the next world, consisting of four islands: Boat A always leads back to the starting island while boat B brings the user one island further. This finite state machine represents a regular language where all words of the language end with at least three Bs. This language needs to be understood unless the player wants to spend a certain amount of time with trial and error approaches. Again, after completing all routes and opening the treasure chest, the learner gets back to the main island where Pengu explains the next finite state machine. In the last level, the islands represent states from a coffee vending machine (nothing inserted, 50ct inserted, 1 Euro inserted, coffee) and the boats represent the possible inputs (50ct, 1 Euro, Cancel). This time, the student has to find all twelve routes to open the treasure chest. After returning to the main island, Pengu complains that the player stole his rent and congratulates him/her for being a real pirate.

The interaction was, again, gaze-based via point-and-click (selecting boats, speaking to the penguin, opening the treasure chest). In *Pengu's Treasure Hunt*, the player changes scenes by selecting boats. Possible islands are: an island with palms, an island with penguins and snow, an island with a shipwreck, and an island with a wood structure. In the laptop and the mobile VR setting, the map is shown when the player looks down. Regarding the HTC Vive setting, the map is attached to one of the controllers. In all settings, the map is extended with every discovered island and route.

4 METHOD

For evaluating the effectiveness of the CSR environments, we conducted a study comparing pre- and posttest performance of students using the EVEs presented in section 3. In order to determine the role of the provided level of immersion for learning processes, we used different technologies for the EVEs: a laptop, a mobile VR, and a HTC Vive were used in the study. The hypotheses deriving from the theoretical background presented in section 2 are:

- (1) The CSR environments are effective for teaching the related CSE contents.
 - (a) The *components of a computer* post-test performance is significantly higher than the pre-test performance after completing the EVE "Bill's Computer Workshop".
 - (b) The *asymmetric encryption* post-test performance is significantly higher than the pre-test performance after completing the EVE "Fluxi's Cryptic Potions".
 - (c) The *finite state machines* post-test performance is significantly higher than the pre-test performance after completing the EVE "Pengu's Treasure Hunt".
- (2) The learning outcomes are higher in the immersive setting than in the non-immersive setting.
 - (a) The mean *components of a computer* post-test performance of students who used the HTC Vive setting is significantly higher than the post-test performance of students who used the laptop setting.
 - (b) The *asymmetric encryption* post-test performance of students who used the HTC Vive setting is significantly higher than the post-test performance of students who used the laptop setting.
 - (c) The *finite state machines* post-test performance of students who used the HTC Vive setting is significantly higher than the post-test performance of students who used the laptop setting.
- (3) There is a significant, positive effect of the level of immersion on the standardized post-test performance over all environments.

4.1 Sample and Procedure

78 (36 f, 4 missing) students from an austrian school took part in the study. The participants were between 13 and 16 years old ($M = 13.95$, $SD = .74$); they went to eighth grade (41) or ninth grade (33) (4 missing).

Table 1: Technological Settings for the Groups

	Group A	Group B	Group C
Bill's Computer Workshop	Mobile VR	HTC Vive	Laptop
Fluxi's Cryptic Potions	HTC Vive	Laptop	Mobile VR
Pengu's Treasure Hunt	Laptop	Mobile VR	HTC Vive

One week prior to the study, the participants filled out the pre-test for the three different topics (see section 4.2). For the study, six students at a time were assigned to two hour slots. Every group played all three EVEs but with varying technologies. Depending on the group, they were provided with different immersive settings

(tab. 1): a laptop, a mobile VR (Daydream View with a Moto Z smartphone), and a professional HMD (HTC Vive). Three participants could play the EVEs at a time. After the experience, the three participants switched rooms and filled out a presence questionnaire (which is not part of this study) and the post-test in the assessment room while the three waiting participants in turn switched to the treatment room (both rooms were supervised by the investigator's team). This procedure was repeated until all participants of a group played all EVEs and filled out the corresponding post-tests. Neither the time for the VR experience nor the time for answering the questionnaires was limited (though no student took longer than 20 minutes for an experience or a questionnaire). Three students dropped out because of motion sickness, another three students did not finish the last environment due to their class schedule.

4.2 Instruments

Like the learning objectives of the particular EVEs (see section 3), the tasks of the pre- and post-test were related to the cognitive classes *knowledge*, *comprehension*, and *application*. Further, the tasks of the pre- and of the post-test were the same. As the answers could either be right or wrong with a sample solution (one point for every correct allocation, key word, or arrow), no second rater was consulted. The students were thought of to have no to poor previous knowledge which is why only the post-tests scale reliabilities are reported. The tasks were designed so that they were solvable without specific knowledge from the environments as they provided brief explanations of the assessed concepts (for the asymmetric encryption/decryption and finite state machines tasks).

The test for the *components of a computer* was related to the cognitive class of *knowledge* and asked the student to insert the given components of a computer (input devices, output devices, input-output-control/motherboard, processor/CPU, secondary memory, primary memory/RAM) in the blanks of a displayed conceptual model (six points). The second task asked the student to name the functions of the components processor/CPU, secondary memory, and primary memory/RAM (three points). Hence, for succeeding in the test, the students had to remember the scheme displayed in the environment and the named functions of the components. The post-test's Cronbach's Alpha value ($\alpha = .16$) was alarming. As the *components of a computer* test consisted of two separate *remembering* tasks referring to different contents rather than a concept that had to be understood, it is possible to argue that students who spent more time searching for the items and exploring the world remembered the functions of the components better while students who spent more time talking to the NPC remembered the scheme better. Following this argumentation, the poor scale reliability can be explained in order that both tasks, while referring to different contents (relations and functions), contributed to the *knowledge* about the components of a computer.

The *asymmetric encryption/decryption* test was applicable for the cognitive class *comprehension*. The tasks assessed the student's understanding skills in terms of the public/private key encryption and decryption process. The first task described a scenario where Alice encrypts a message with Bob's public key so that only Bob can read its content and asked the student to explain why (two points). The second task asked the student to reason why Alice

would use her own private key in order to send a signed message which can be encrypted by everyone but, in turn, everyone can be sure that Alice sent the message (two points). In the third task, the student had to use the right keys to encrypt messages decrypted with given keys (four points). Again, the post-test's scale reliability was poor ($\alpha = .61$). When the third task was removed, the Cronbach's Alpha increased to $\alpha = .68$. Even though the third task was designed to contribute to the *comprehension* class as the student had to understand which key had to be used rather than performing an encryption/decryption process by him-/herself, it seemed to be associated with the *application* class more which is why it was removed to increase the test's reliability.

For assessing the students' *application* skills for the *finite state machines* topic, the three tasks of the test related to the metaphor used in the corresponding EVE where states were displayed as islands, inputs as boats, transition functions as boat routes between the islands and final states as treasure islands. For the first task (seven points), the students had to draw a map according to a given description (e.g. *Coming from the desert island, boat A leads to the palm island and boat B leads to the ice island.*). For the second task, a text described the function of the finite state machine: On every island, one boat should lead to another island while the other boat leads back directly to the same island. Given this description, the students had to fill out blanks in a gap text (e.g. *Coming from the shipwreck island, boat A leads to ...*) to create a finite state machine that fulfills the given requirements (seven points). For the third task, students had to draw a complete finite state machine (an island map where every island has a boat A and a boat B) on their own (seven points). The description of the map was as follows: In this island world, the treasure is hidden on pirate island and the starting island is palm island. Draw the routes of the boats so that the treasure can only be found if the last two boats taken were boat B. Further, the students were given some example routes that could lead to the treasure island (e.g. BB, ABB, ABABB, etc.). The Cronbach's Alpha value was questionable ($\alpha = .54$). The test was maintained the way it was designed, but the reliability value has to be kept in mind for further discussions about the results from this test.

4.3 Findings

Table 2 shows the means and the standard deviations for the pre-tests and for the post-tests. For the *components of a computer* assessment, a mean learning success of 29% derived from a significant difference [$T(64) = 8.87, p < .0005$] between the pre- and the post-test. A significant difference [$T(64) = 3.27, p < .01$] between the *asymmetric encryption/decryption* pre- and post-test revealed a mean learning success of 13%. The mean performance gain of 12% regarding the *finite state machines* pre- and posttests was significant as well [$T(65) = 4.31, p < .0005$].

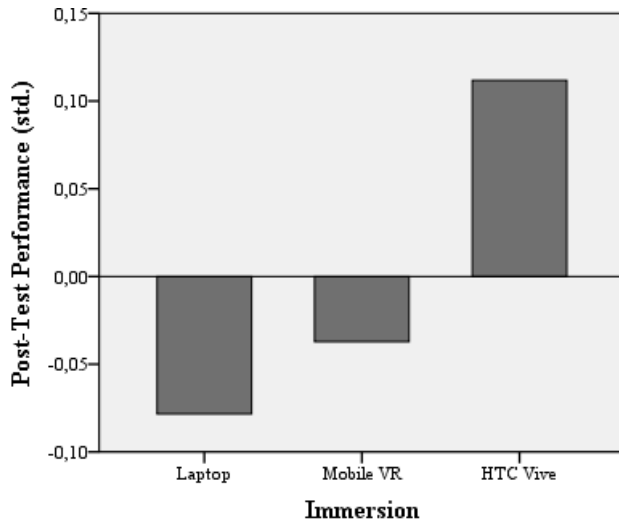
For comparing immersive vs. non-immersive settings, the post-test results of the HTC Vive group were compared to the post-test results of the laptop group (the groups varied for each environment). Levene's tests were insignificant for all paired groups, indicating equal variances. The *components of a computer* post-test-scores for the HTC Vive group ($M = 5.93, SD = 2.57$) were higher than for the laptop group ($M = 5.00, SD = 2.27$); the result was not significant

Table 2: Results from the Pre-Test and from the Post-Test

	M_{pre}	SD_{pre}	M_{post}	SD_{post}
CC ¹	3.28	2.04	5.86	2.36
AED ²	1.32	1.16	1.83	1.20
FSM ³	9.89	2.89	12.30	3.70

Note. ¹Components of a Computer; ²Asymmetric Encryption/Decryption; ³Finite State Machines

[$T(47) = 1.34, p > .05, d = .38$]. The scores for the *asymmetric encryption/decryption* post-test were higher for HTC Vive group ($M = 2.13, SD = 1.14$) than for the laptop group ($M = 1.06, SD = 1.14$) as well. The difference was not significant with $T(49) = 1.47, p > .05, d = .41$. For the *finite state machines* post-test, the HTC Vive group ($M = 12.55, SD = 3.83$) scored slightly lower than the laptop group ($M = 13.18, SD = 3.83$). Again, the differences were not significant; $T(40) = .54, p > .05, d = -.17$.

**Figure 5: Standardized Post-Test Results for the Different Levels of Immersion**

For an ANOVA comparing the the post-test results for the three levels of immersion (see Fig. 5), the test scores were z-standardised. The analysis of variance for the standardised post-test results between the laptop setting ($M = -.08, SD = 1.00$), the mobile VR setting ($M = -.04, SD = 1.02$), and the HTC Vive setting ($M = .11, SD = .97$) showed a small, non-significant effect of the level of immersion on post-test results, $F(2, 208) = .72, p > .05, \eta_p^2 = .01$.

4.4 Discussion and Limitations

Our results show that the designed EVEs are effective for teaching the contents *components of a computer*, *asymmetric encryption/decryption*, and *finite state machines*. By using (metaphorical) narratives to explain the ideas of *asymmetric encryption/decryption*

and *finite state machines*, as well as presenting selected components of computing systems, our game-based learning environments could foster the students' *knowledge, comprehension*, and *application* skills. By maintaining the sub-hypotheses H_{1a} – H_{1c} , we can maintain H_1 : *The CSR environments are effective for teaching the related CSE contents*. Of course, the students did not show perfect scores after completing the environments. It is recommended that the EVEs are used as a motivation to start the particular topic and then analyze the experiences together with a teacher. We are aware that this study lacks a control group without any treatment. As the focus of this study was set on the comparison of different technologies this is a factor that has to be considered in future studies. Further, the presented study can not compare the results from the CSR approaches to other ways of teaching these concepts, like CSU activities or more traditional teaching methods. Hence, we do not propose the CSR method as being superior to other teaching methods but rather present CSR as a new way introduce complex CS contents in a motivating way using virtual hands-on approaches.

Regarding H_2 : *The learning outcomes are higher in the immersive setting than in the non-immersive setting*, the results are incoherent: While the *components of a computer* and the *asymmetric encryption/decryption* topics could benefit from more immersive technology, the *finite state machines* learning process seemed to be hindered by the use of VR. It could be argued that the interaction might have been distracting as *Pengu's Treasure Hunt* involved many scene changes and exploring the environment did not contribute to the learning process. But also, the poor scale reliability of the *finite state machines* test has to be kept in mind. Further investigations should reflect on the validity of the presented test; maintaining the island metaphor in a test might not be suitable to assess skills regarding the *application of finite state machine concepts*. While we have to discard H_2 and its sub-hypotheses H_{2a} – H_{2c} for now due to the non-significant results, *Bill's Computer Workshop* and *Fluxi's Cryptic Potions* give us indicators that the CSR approach might benefit from immersive technologies. To investigate this idea further, larger sample sizes are needed.

Similar to H_2 , the results for H_3 show indicators that the level of immersion might have a positive effect on the learning outcomes. As these effects are not significant, we can not maintain H_3 for now. Here too, larger sample sizes would be needed in order to investigate this effect further. As the effect sizes on learning outcomes for using non-immersive versions vs. immersive versions of the particular environments vary as well, it is possible that immersion is not be the crucial factor for learning processes. Several approaches suggest that, instead of focusing on the technological features of VR, investigating the effect of presence, as the subjective feeling deriving from these features, might be expedient. This idea is also compatible to the original CSU approach: A more 'real' (perceived as non-mediated) hands-on experience enhances motivation and engagement and therefore, fosters students' Computational Thinking skills by actively becoming part of the activity.

While the first approaches to evaluate *Computer Science Re-plugged* show that immersive technology can contribute to learning processes related to CS Education, additional studies have to be carried out in the future. In particular, the level of immersion seems to influence the level of presence in the environments while not being the main source for learning activities.

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